PSC I-Girder Bridge Design and Analysis Report

New PSC Bridge Project
Prepared by: Ranvir Kumar (23CE36016) Date: 31 October 2026
Design basis: IRC:6-2017 / IRC:112-2011 (SP:71 as detailing reference)

1 of 14 checks FAIL

1. Description

Design of a simply supported post-tensioned PSC I-girder bridge for CE60258.

The structure is a simply supported, post-tensioned prestressed concrete (PSC) I-girder bridge with an effective span of 33.00 m. The deck is 14.50 m wide and carries 3 traffic lanes of 3.5 m (10.50 m carriageway), a 1.5 m footpath and a 0.5 m crash barrier on each side. It is supported on 5 precast I-girders at 2.90 m spacing, each 2.00 m deep, carrying a 250 mm cast-in-situ RCC deck slab and 5 cross diaphragms.

Girders use M50 concrete, Fe500 reinforcement and 1860 MPa strands. Prestress is provided by 3 post-tensioned cables of 17 strands (7,140 mm²), sized automatically, jacked to 1395 MPa (P0 = 9,960 kN). After friction, anchorage slip, elastic shortening, creep, shrinkage and relaxation the effective force at midspan is 7,981 kN, a total loss of 19.9 %.

Live load is taken as Class 70R Wheeled with a worst-girder distribution factor of 0.462 and an impact factor of 0.115. The ultimate design moment at midspan is 16,032 kNm against a capacity of 19,939 kNm; the critical design shear is 1,348 kN against 1,561 kN. The long-term midspan deflection is 28.1 mm (limit 132 mm).

Of 14 design checks, 13 pass and 1 fail: Transfer: max compression. The failing check(s) need a design change (for example staged stressing, a higher strength at transfer, or a larger bottom flange).

Design moment Mu	Moment capacity	Design shear	Pe at midspan	Total deflection
16,032 kNm	**19,939 kNm**	**1,975 kN**	**7,981 kN**	**28.1 mm**

2. Input Data

Span and deck	
Effective span	**33.00 m**
Deck width	**14.50 m**
Carriageway	**10.50 m**
Lanes	**3 x 3.5 m**
Footpath / barrier	**1.5 m / 0.5 m**
Girders / spacing	**5 @ 2.90 m**
Overhang	**1.45 m**
Slab / haunch	**250 mm / 50 mm**
Diaphragms	**5 x 300 mm**

Girder section	
Depth	**2000 mm**
Top flange	**1000 x 200 mm**
Web	**300 mm**
Bottom flange	**800 x 300 mm**
Area A	**0.8030 m²**
Centroid from soffit	**997 mm**
Inertia I	**0.3676 m⁴**
Zb / Zt	**0.3688 / 0.3664 m³**
Composite Ic	**0.8869 m⁴**

Materials	
Girder concrete fck	**50 MPa**
Deck concrete fck	**35 MPa**
Ecm (girder)	**37,659 MPa**
Reinforcement fy	**500 MPa**
Strand fpk	**1860 MPa**
Strand area	**140 mm²**

Prestressing	
Cables x strands	**3 x 17**
Aps	**7,140 mm²**
Jacking stress	**1395 MPa**
μ / k	**0.17 / 0.0015**
Anchorage draw-in	**6.0 mm**
Creep φ / shrinkage	**1.8 / 3.5e-04**
Relaxation (1000 h)	**2.5 %**

Loads	
Self weight + slab	**45.05 kN/m**
SIDL	**7.75 kN/m**
Wearing coat	**3.00 kN/m**
Footpath live	**2.40 kN/m**
Governing vehicle	**Class 70R Wheeled**
Distribution factor	**0.462**
Impact factor	**0.115**

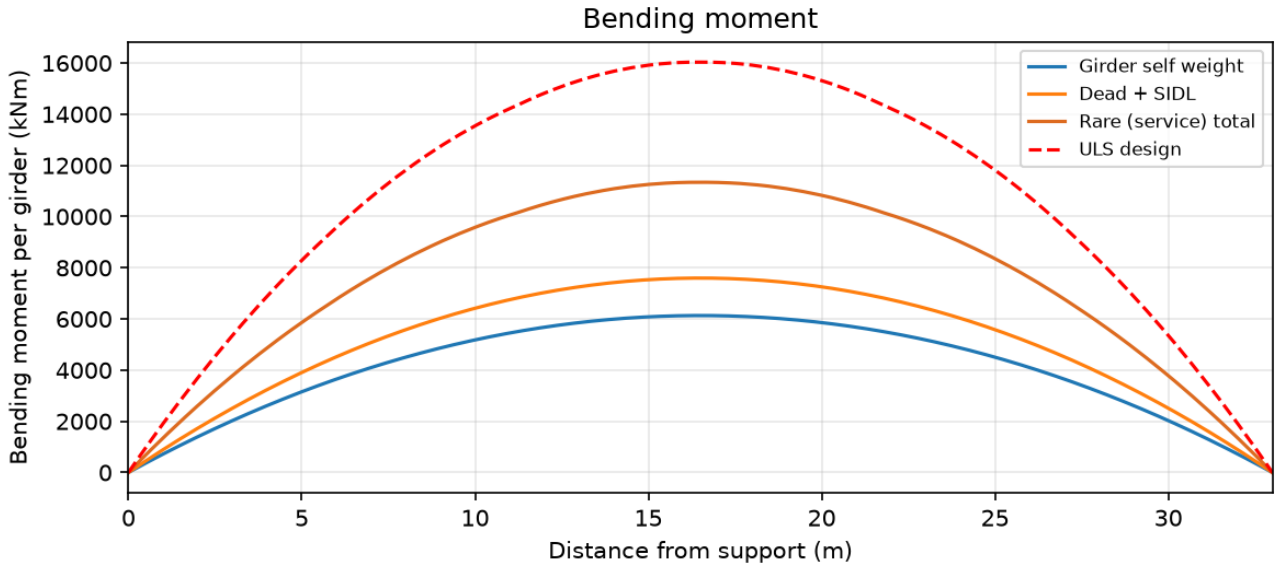
3. Design Checks Summary

Check	Demand	Limit	Unit	Util.	Status	Clause
Flexural capacity (ULS)	16,032.32	19,938.62	kNm	80.4 %	PASS	IRC:112 Sec.8 (verify)
Shear capacity (ULS)	1,347.82	1,560.84	kN	86.4 %	PASS	IRC:112 cl.10.3 (verify)
Web crushing VRd,max	1,347.82	1,560.84	kN	86.4 %	PASS	IRC:112 cl.10.3.3 (verify)
Transfer: max compression	22.78	22.50	MPa	101.3 %	FAIL	0.6 fck(t), EN 1992 5.10.2.2 / IRC:112 (verify)
Transfer: max tension	0.00	3.36	MPa	0.0 %	PASS	tension limit = fctm(fci) at transfer (adopted)
Service rare: max compression	12.54	24.00	MPa	52.3 %	PASS	IRC:112 cl.12.2.1 (verify)
Service rare: bottom tension	0.00	0.00	MPa	-	PASS	decompression/tension limit (adopted)
Service QP: max compression	10.64	18.00	MPa	59.1 %	PASS	IRC:112 cl.12.2.1 (verify)
Service QP: bottom decompression	0.00	0.00	MPa	-	PASS	IRC:112 Table 12.1 (verify)
Live-load deflection	12.74	41.25	mm	30.9 %	PASS	L/800 (verify clause)
Total deflection (long-term)	28.08	132.00	mm	21.3 %	PASS	L/250 (verify clause)
Prestress: jacking stress	1,395.00	1,473.12	MPa	94.7 %	PASS	IRC:112 cl.12.2.2 (verify)
Prestress: Pe < P0	7,980.97	9,960.30	kN	80.1 %	PASS	IRC brief
Cables fit in bottom flange / web	0.47	1.00	m	47.7 %	PASS	all ducts below centroid; web ≥ duct + 100 mm (IRC:112 cl.15, verify)

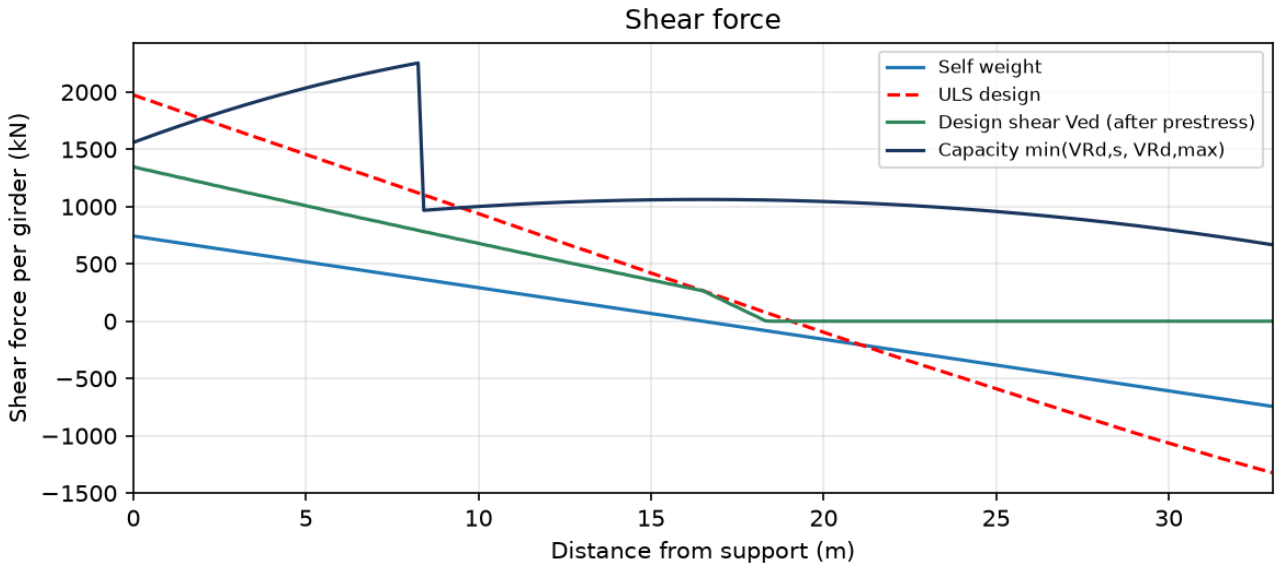
Utilisation = demand / limit. Clause references marked (verify) must be confirmed against the current code editions.

4. Diagrams

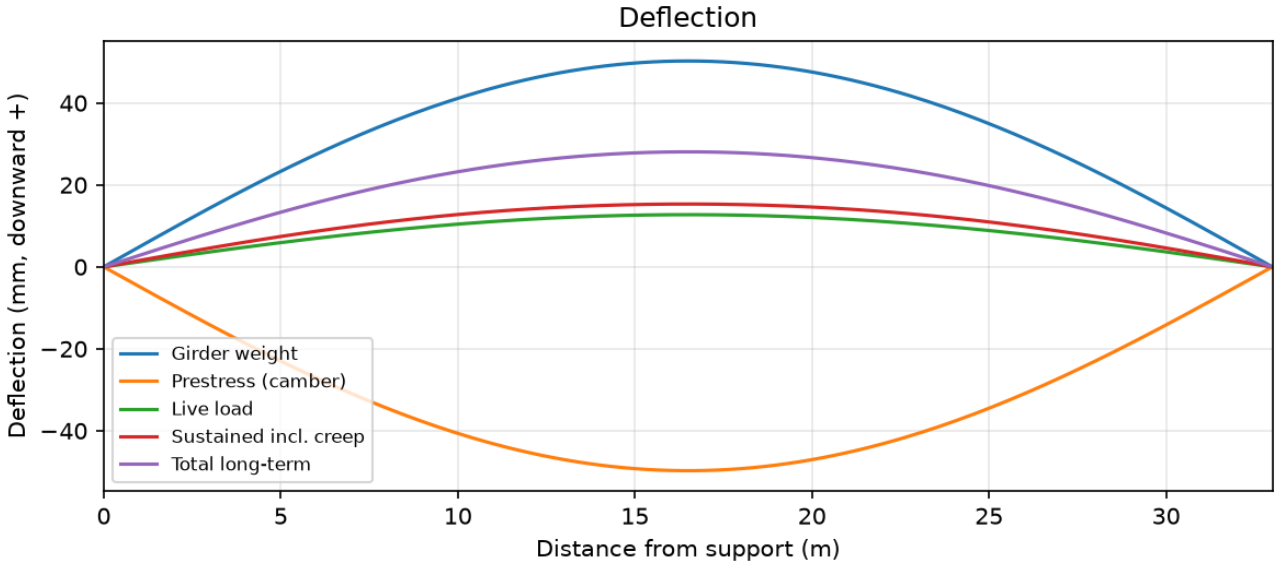
Bending moment



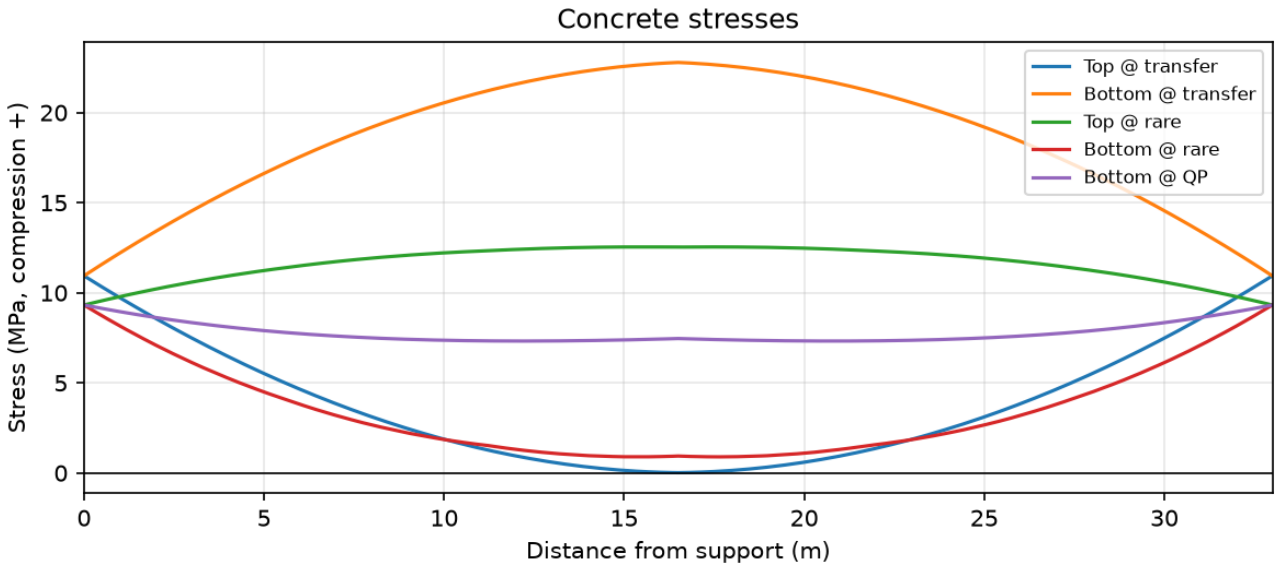
Shear force



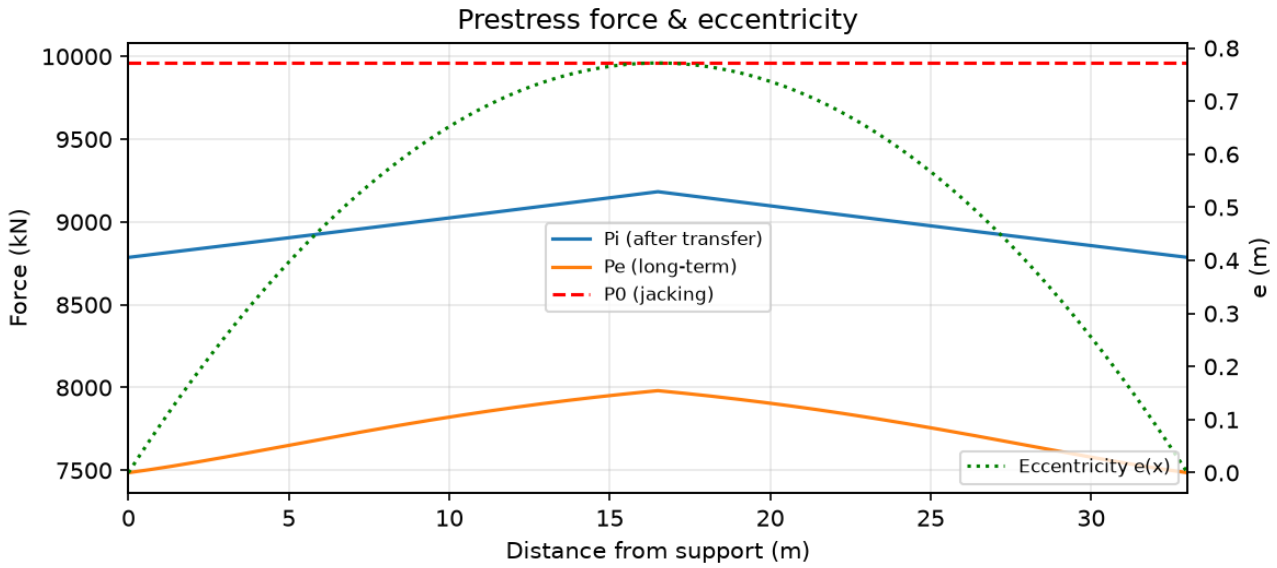
Deflection



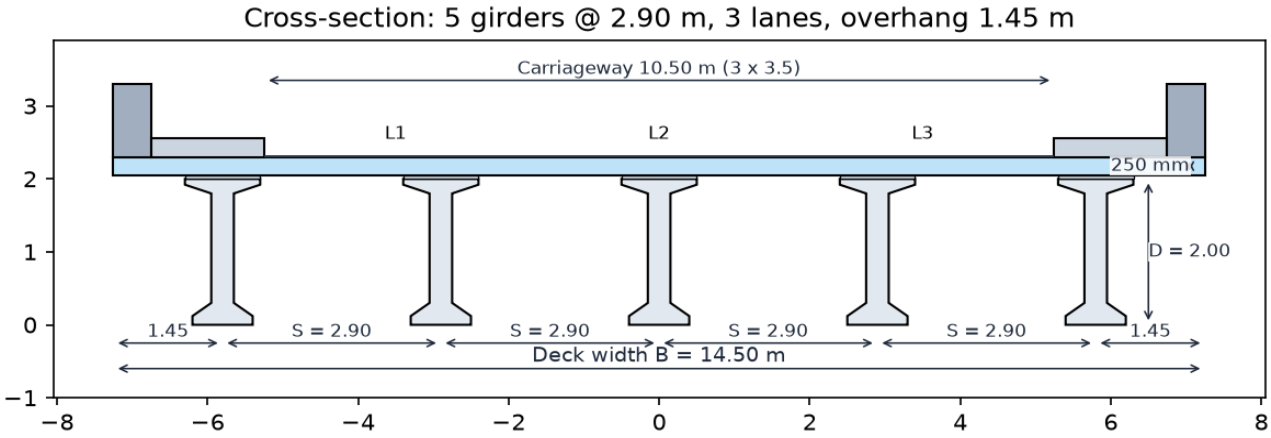
Concrete stresses

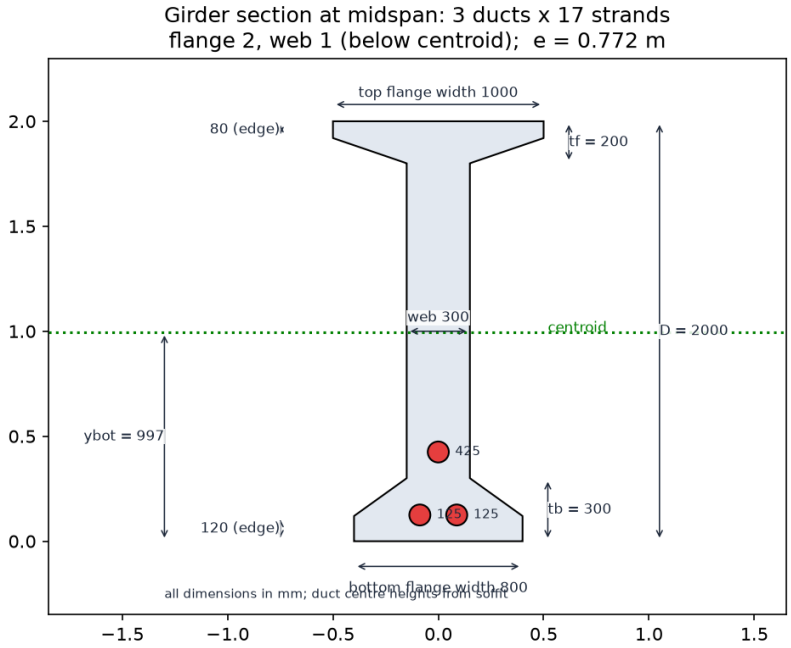


Prestress force & eccentricity



Bridge cross-section and girder section





5. Step-by-Step Calculations

Each check is shown as: Formula (blue italics), Substitution with the actual values (monospace) and Result (bold). Values are in kN, m and MPa unless stated.

1. Flexural capacity (ULS) — PASS

Demand	Limit	Utilisation	Clause / location
16,032.32 kNm	**19,938.62 kNm**	**80.4 %**	IRC:112 Sec.8 (verify) [midspan]

1. Design moment at midspan

Formula	*Mu = 1.35 (Mg + Msidl) + 1.75 Mwc + 1.5 (Mll + Mfp)*
Values	`Mu = 1.35 × (6132.7 + 1055.0) + 1.75 × 408.8 + 1.5 × (3415.7 + 326.7)`
Result	**Mu = 16,032.3 kNm**
Note	Live load: Class 70R Wheeled; distribution factor 0.462; impact 0.115

2. Moment capacity (strain compatibility, rectangular stress block)

Note	λ = 0.8, fcd = 0.67 fck / 1.5, fpd = 0.88 fpk / 1.15
Values	`d = 2.075 m; Aps = 7,140 mm²; fpd = 1423.3 MPa; pre-strain εp0 = 0.00634`
Values	`Neutral axis x = 0.281 m (x/d = 0.136)`
Values	`Tendon stress = 1423.3 MPa → T = 10,162 kN; C = 10,246 kN`
Formula	*Mu,cap = T × z*
Values	`Mu,cap = 10,162 × 1.962`
Result	**Mu,cap = 19,938.6 kNm**

3. Check

Result	**Mu = 16,032.3 kNm ≤ Mu,cap = 19,938.6 kNm → utilisation 80.4 %**

2. Shear capacity (ULS) — PASS

Demand	Limit	Utilisation	Clause / location
1,347.82 kN	**1,560.84 kN**	**86.4 %**	IRC:112 cl.10.3 (verify) [x = 0.00 m]

1. Design shear at x = 0.00 m

Formula	*VEd = V_uls − 0.9 Vp, V_uls = 1.35 (Vg + Vsidl) + 1.75 Vwc + 1.5 (Vll + Vfp)*
Values	`V_uls = 1,975.3 kN; Vp = Pe sin θ = 697.2 kN`
Result	**VEd = 1,975.3 − 0.9 × 697.2 = 1,347.8 kN**

2. Section and links

Values	`d = 1.303 m; z = 0.9 d = 1.173 m; bw = 300 mm; cot θ = 2.5`
Values	`Links: 2-leg 10 mm @ 125 mm (Asw = 157.1 mm²); fywd = fy / 1.15 = 434.8 MPa`

3. Capacity

Formula	*VRd,s = (Asw / s) · z · fywd · cot θ*
Result	**VRd,s = 1,602.1 kN (VRd,c without links = 534.9 kN, for information)**
Result	**Capacity = min(VRd,s ; VRd,max) = 1,560.8 kN**

4. Check

Result	**VEd = 1,347.8 kN ≤ 1,560.8 kN → utilisation 86.4 %**

3. Web crushing VRd,max — PASS

Demand	Limit	Utilisation	Clause / location
1,347.82 kN	**1,560.84 kN**	**86.4 %**	IRC:112 cl.10.3.3 (verify) [x = 0.00 m]

1. Strut parameters

Formula: *ν1 = 0.6 (1 − fck / 250); αcw = 1 + σcp / fcd*

Values: `ν1 = 0.480; fcd = 22.33 MPa; σcp = 4.47 MPa; αcw = 1.200`

2. Strut capacity

Formula: *VRd,max = αcw · bw · z · ν1 · fcd / (cot θ + tan θ)*

Values: `VRd,max = 1.200 × 0.30 × 1.173 × 0.480 × 22.33 × 10³ / (2.5 + 0.40)`

Result: **VRd,max = 1,560.8 kN**

3. Check

Result: **VEd = 1,347.8 kN ≤ 1,560.8 kN → utilisation 86.4 %**

4. Transfer: max compression — FAIL

Demand	Limit	Utilisation	Clause / location
22.78 MPa	**22.50 MPa**	**101.3 %**	0.6 fck(t), EN 1992 5.10.2.2 / IRC:112 (verify)

1. Stage and strength

Note: Girder alone: initial prestress Pi plus girder self weight only.

Values: `fci = 0.75 × 50 = 37.5 MPa`

2. Stress at the governing section

Formula: *σ = Pi / A ± Pi · e / Z ∓ Mg1 / Z*

Values: `x = 16.50 m; Pi = 9,183 kN; e = 0.772 m; Mg1 = 2,903 kNm`

Result: **σtop = 0.02 MPa; σbottom = 22.78 MPa**

3. Check

Result: **σmax = 22.78 MPa vs limit 0.6 fci = 22.50 MPa**

Note: Limit adopted from EN 1992-1-1 cl. 5.10.2.2; confirm the IRC:112 value.

5. Transfer: max tension — PASS

Demand	Limit	Utilisation	Clause / location
0.00 MPa	**3.36 MPa**	**0.0 %**	tension limit = fctm(fci) at transfer (adopted)

1. Governing section

Values: `x = 16.50 m: σtop = 0.02 MPa; σbottom = 22.78 MPa (negative = tension)`

2. Limit

Formula: *fctm(fci) = 0.3 · fci^(2/3)*

Result: **fctm = 3.36 MPa (adopted)**

6. Service rare: max compression — PASS

Demand	Limit	Utilisation	Clause / location
12.54 MPa	**24.00 MPa**	**52.3 %**	IRC:112 cl.12.2.1 (verify)

Note: Stages: Pe and girder loads on the girder alone; SIDL and live load on the composite section.

Values: `Maximum girder top stress = 12.54 MPa; maximum deck slab stress = 4.06 MPa`

Formula: *Limit = 0.48 fck*

Result: **Limit = 24.00 MPa**

7. Service rare: bottom tension — PASS

Demand	Limit	Utilisation	Clause / location
0.00 MPa	**0.00 MPa**	**-**	decompression/tension limit (adopted) [x = 15.35 m]

Formula	*σbottom = Pe / A + Pe · e / Zb − Mg / Zb − Mcomp · ybc / Ic*
Values	`x = 15.35 m; Pe = 7,958 kN; e = 0.768 m; Mcomp = 5,189 kNm`
Result	**σbottom = 0.89 MPa (negative = tension); permitted tension = 0.00 MPa**
Note	Automatic sizing selected 3 cable(s) × 17 strands to satisfy this check at midspan.

8. Service QP: max compression — PASS

Demand	Limit	Utilisation	Clause / location
10.64 MPa	**18.00 MPa**	**59.1 %**	IRC:112 cl.12.2.1 (verify)

Values	`Maximum girder top stress = 10.64 MPa; maximum deck slab stress = 1.14 MPa`
Formula	*Limit = 0.36 fck*
Result	**Limit = 18.00 MPa**

9. Service QP: bottom decompression — PASS

Demand	Limit	Utilisation	Clause / location
0.00 MPa	**0.00 MPa**	**-**	IRC:112 Table 12.1 (verify)

Values	`Minimum bottom stress under quasi-permanent load = 7.32 MPa`
Result	**Requirement: σbottom ≥ 0 (no decompression)**

10. Live-load deflection — PASS

Demand	Limit	Utilisation	Clause / location
12.74 mm	**41.25 mm**	**30.9 %**	L/800 (verify clause)

Note	Double integration of M / EI on the composite section; simply supported, zero deflection at both supports.
Values	`EIc = Ecm × Ic = 37,658,937 kN/m² × 0.8869 m⁴`
Result	**δLL (midspan) = 12.74 mm**
Formula	*Limit = L / 800*
Result	**Limit = 41.25 mm**

11. Total deflection (long-term) — PASS

Demand	Limit	Utilisation	Clause / location
28.08 mm	**132.00 mm**	**21.3 %**	L/250 (verify clause)

Values	`Girder weight = 50.25 mm; prestress = -49.74 mm; SIDL = 4.97 mm (downward positive)`
Formula	*δsustained = (1 + φ) · (δg + δsidl + δp)*
Values	`δsustained = (1 + 1.8) × 5.48 = 15.34 mm`
Result	**δtotal = δsustained + δLL = 28.08 mm**
Formula	*Limit = L / 250*
Result	**Limit = 132.0 mm**

12. Prestress: jacking stress — PASS

Demand	Limit	Utilisation	Clause / location
1,395.00 MPa	**1,473.12 MPa**	**94.7 %**	IRC:112 cl.12.2.2 (verify)

Formula	*σj = ratio × fpk*
Values	`σj = 0.75 × 1860 = 1395 MPa`
Formula	*Limit = min(0.8 fpk ; 0.9 fp0.1k)*
Result	**Limit = min(1488 ; 1473) = 1473 MPa**

13. Prestress: Pe < P0 — PASS

Demand	Limit	Utilisation	Clause / location
7,980.97 kN	**9,960.30 kN**	**80.1 %**	IRC brief

Formula	*P0 = σj × Aps*
Values	`P0 = 1395 MPa × 7,140 mm² = 9,960 kN`
Note	Losses: friction (μ = 0.17, k = 0.0015/m), anchorage draw-in 6.0 mm, elastic shortening, creep, shrinkage, relaxation.
Values	`At midspan: Pi = 9,183 kN; Pe = 7,981 kN`
Result	**Total loss = 19.9 % of P0**

14. Cables fit in bottom flange / web — PASS

Demand	Limit	Utilisation	Clause / location
0.47 m	**1.00 m**	**47.7 %**	all ducts below centroid; web ≥ duct + 100 mm (IRC:112 cl.15, verify)

1. Duct layout

Values	`3 ducts of Ø100 mm: 2 in the bottom flange, 1 stacked in the web (below the centroid)`
Values	`Top of duct group = 0.475 m above soffit; girder centroid = 0.997 m`
Result	**All ducts below the centroid: yes**

2. Web width (needed only if ducts are placed in the web)

Formula	*Required web width ≥ duct diameter + 2 × 50 mm*
Result	**Required = 200 mm; provided = 300 mm**

6. Calculation Notes (global)

Section properties (girder alone)

Given	D=2.00 m, bt=1.00, tf=0.20, tw=0.30, bb=0.80, tb=0.30 m
Formula	*A = Σ b·Δy ; ȳ = Σ b·y·Δy / A ; I = Σ b(y-ȳ)²Δy*
Values	`A=0.8030 m², ȳb=0.997 m, I=0.3676 m⁴`
Result	**Zb=0.3688 m³, Zt=0.3664 m³**
Check	numerical integration, dy=2 mm

Dead load per girder

Given	slab 0.25 m, deck width W=14.50 m, 5 girders
Formula	*w = A·25 + haunch + W·t_slab·25/n + diaphragms*
Values	`w = 21.32 + 23.73`
Result	**45.05 kN/m**
Check	M_DL(mid) = wL²/8 = 6132.7 kNm

Live load

Given	governing vehicle Class 70R Wheeled, DF=0.462 (1 lane(s), Courbon), impact=11.5%
Formula	*M_LL = DF · (1+i) · M_vehicle,max (moving-load envelope)*
Values	`M_LL(mid) = 3415.7 kNm incl. impact`
Result	**3415.7 kNm**
Check	envelope from 0.1 m load steps, both directions

Prestress force and losses (midspan)

Given	3 cables x 17 strands, Aps=7140 mm², σj=1395 MPa
Formula	*P0 = σj·Aps ; Pe = P0 - friction - slip - ES - (creep+shrinkage+relaxation)*
Values	`P0=9960 kN ; Pi(mid)=9183 kN ; Pe(mid)=7981 kN`
Result	**total loss 19.9 %**
Check	Pe < P0

Flexural capacity (ULS)

Given	d=2.075 m, Aps=7140 mm², fpd=1423 MPa, x=0.281 m
Formula	*T = σp·Aps = C ; Mu,cap = T·(d - ȳc)*
Values	`T=10162 kN, lever arm=1.962 m`
Result	**Mu,cap = 19939 kNm**
Check	Demand Mu = 16032 kNm

Shear (variable-strut method)

Given	critical x=0.00 m, bw=0.30 m, z=1.17 m, cotθ=2.5
Formula	*VRd,s = (Asw/s)·z·fywd·cotθ ; VRd,max = αcw·bw·z·ν1·fcd/(cotθ+tanθ)*
Values	`VRd,s=1602 kN, VRd,max=1561 kN`
Result	**Ved = 1348 kN ; 2L-10mm @ 125/300 mm**
Check	utilisation 0.86

7. Assumptions to Verify

- Courbon (rigid cross-beam) distribution of live load to the worst girder; no grillage/FE refinement.
- IRC:6 lane reduction taken as 1.0 (<=2 lanes), 0.9 (3), 0.8 (>=4) - verify against IRC:6-2017.
- 70R wheeled/tracked treated as a single lane vehicle; Class A/B may fill all lanes. 'Governing' envelopes all.
- Impact factors: Class A/B 4.5/(6+L); 70R wheeled 25% to 12 m then 4.5/(6+L); tracked 25%->10% - verify.
- ULS partial factors: DL & SIDL 1.35, wearing coat 1.75, live 1.5 - verify IRC:6-2017 Table B.2.
- Deck cast on unpropped girders: girder alone carries self-weight, slab, diaphragms; composite carries SIDL and LL.
- SIDL shared equally by all girders (no outer-girder cantilever effect).
- Wind, temperature, braking, seismic are NOT modelled (the GUI no longer invents values for them).
- Time-dependent losses via the EN 1992-1-1 eq. 5.46 form; shrinkage/creep inputs are user values.
- Flexure: strain compatibility, rectangular block 0.8x, fcd = 0.67fck/1.5, strand fpd = 0.88fpk/1.15.
- Shear: variable-strut method with ν1 = 0.6(1-fck/250); fywd = fy/1.15 - verify IRC:112 cl.10.3.
- End-block bursting/anchorage-zone design, interface shear, diaphragm design and handling are NOT checked.
- Section is prismatic except optional end-block web thickness (used for shear only).